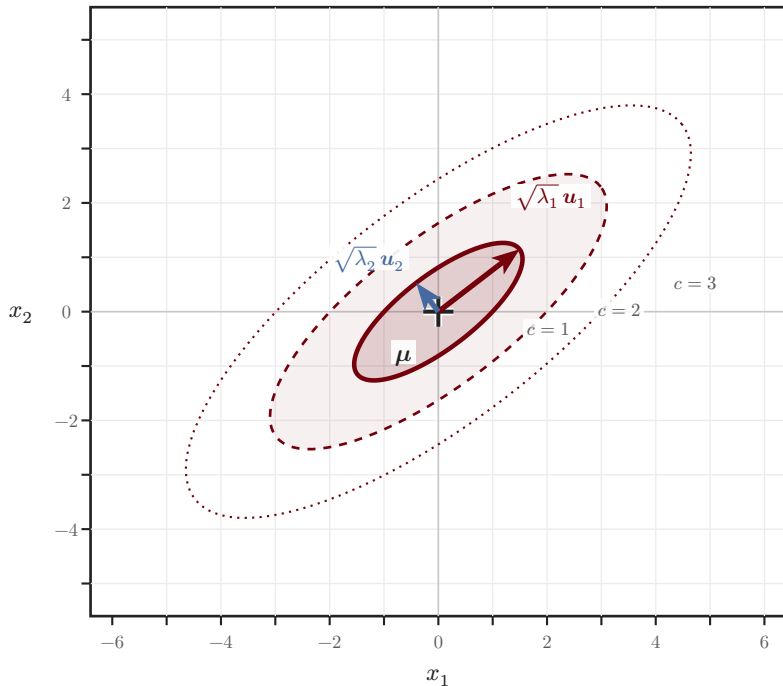


Gaussian covariance contours

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$$

iso-probability ellipses of $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$



$$\boldsymbol{\Sigma} = \begin{pmatrix} 2.4 & 1.5 \\ 1.5 & 1.6 \end{pmatrix}$$

$$\boldsymbol{\Sigma} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^\top$$

$$\lambda_1 = 3.55$$

$$\lambda_2 = 0.45$$

$$\text{semi-axes} = c\sqrt{\lambda_i}$$

$$\text{directions} = \mathbf{u}_i$$

$$\text{---} \quad c = 1 (\approx 39\%)$$

$$\text{- - -} \quad c = 2 (\approx 86\%)$$

$$\cdots \quad c = 3 (\approx 99\%)$$